

Technical Report: Indian Food Image Classification Using Transfer Learning

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Abstract

This report details the development and evaluation of an Indian food image classification model. Utilizing a dataset of 8 traditional Indian bread and dish classes, a convolutional neural network was trained using transfer learning on the MobileNetV2 architecture. The model achieved a robust validation accuracy of 92%, demonstrating strong generalization capabilities. Furthermore, the trained model was integrated into a web application designed to provide nutritional and allergen information based on visual input.

1 Introduction

Dietary tracking and automated food recognition are increasingly relevant in health, wellness, and cultural applications. Accurate image-based food classification enables individuals to monitor nutritional intake and avoid potential allergens without manual data entry. Indian cuisine, characterized by a wide variety of visually similar flatbreads and dishes, poses a unique challenge for computer vision systems.

This project aims to build an automated classifier capable of distinguishing between 8 specific Indian food categories. To address the complexities of visual food recognition, we employ transfer learning utilizing the MobileNetV2 architecture. The classification system is additionally deployed via an interactive web application that provides users with predicted dish names, estimated caloric values, and common allergen warnings.

2 Data

The dataset, `gimages_ds`, consists of 1,123 images categorized into 8 distinct classes: *bhature*, *chapati*, *daal_baati_churma*, *lachha_paratha*, *naan*, *palak_roti*, *ragi_roti*, and *rumali_rotti*.

The data was partitioned into an 80-20 split for training and validation, resulting in 899 training images and 224 validation images. To standardize the inputs for the MobileNetV2 architecture, all images were resized to 224×224 pixels and their pixel values were rescaled to the $[-1, 1]$ range.

Given the relatively small size of the dataset, data augmentation techniques were applied exclusively to the training set to prevent overfitting and improve model robustness. The augmentation pipeline included:

- Random horizontal flipping
- Random rotation ($\pm 20\%$)
- Random zooming ($\pm 20\%$)

3 Method

The classification model relies on a transfer learning approach, utilizing the MobileNetV2 base model pre-trained on the ImageNet dataset. The base network's weights were frozen to act as a robust feature extractor, mitigating the risk of destroying pre-learned features during the initial training phase.

The complete network architecture is structured as follows:

1. **Input Preprocessing Layer:** Rescales pixel intensities.
2. **Data Augmentation Layer:** Applies random flips, rotations, and zooms.
3. **Feature Extractor:** Frozen MobileNetV2 base model.
4. **Pooling:** Global Average Pooling 2D layer to reduce spatial dimensions to a 1280-dimensional feature vector.
5. **Regularization:** A Dropout layer with a rate of 0.2 to combat overfitting.
6. **Output Classification:** A Dense layer with 8 units and a softmax activation function to produce class probabilities.

The model was compiled using the Adam optimizer with a learning rate of 0.001 and Sparse Categorical Crossentropy as the loss function. Training was conducted for up to 50 epochs with a batch size of 16. An Early Stopping callback was implemented with a patience of 3 epochs on the validation loss. The callback was configured to restore the best model weights to ensure optimal generalization.

Following training, the model was serialized and embedded into a Streamlit web application. The application maps predictions to a dietary database to alert users of approximate caloric content and allergens (e.g., wheat, dairy, gluten, ghee).

4 Results

The model converged efficiently, with early stopping halting the training process at epoch 30 and restoring the optimal weights from epoch 27. The final restored model achieved a training accuracy of 95.33% and a validation accuracy of 92.41%.

A classification report on the validation set confirms exceptional precision and recall across most categories. The macro-average F1-score stood at 0.92. Notably, the *palak_roti* class achieved perfect prediction metrics (Precision: 1.00, Recall: 1.00). The lowest performance was observed in the *rumali_rotti* class, which still maintained a respectable F1-score of 0.85.

The deployment phase demonstrated that the model's inference time is suitable for real-time web applications, accurately providing end-users with dynamic allergen badges and caloric estimations based on uploaded imagery.

5 Limitations

Despite the high accuracy, several limitations must be acknowledged:

- **Dataset Size:** With only 1,123 images across 8 classes, the dataset is relatively small. The model's ability to generalize to highly varied real-world conditions (e.g., different lighting, partial occlusions, varying plating styles) might be restricted.
- **Intra-class Variance:** Flatbreads such as *chapati* and *rumali_rotti* share high visual similarities. Scaling the model to include more flatbread types (e.g., phulka, tandoori roti) could result in overlapping feature spaces and degraded performance.

- **Fixed Nutritional Database:** The application relies on a hardcoded, static nutritional database. Real-world dishes vary drastically in preparation methods, meaning the caloric and allergen estimations are strictly approximations and cannot account for hidden ingredients or cross-contamination.